FINAL YEAR PROJECT PRESENTATION & DEMO

Resource-Constrained Evaluation of Lightweight Concept Drift Detectors in Machine Learning

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OUTLINE OF PRESENTATION

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Introduction

Problem context: Machine learning models in production face concept drift, where the joint distribution $P(X, y)$ changes over time and degrades prediction quality.

Research gap:

- Existing drift benchmarks usually hold deployment resources constant.
- Real systems must trade off monitoring frequency, detection window size, and reference memory.
- It remains unclear which constraint matters most in practical detection performance.

Study focus: Evaluate lightweight unsupervised detectors under realistic resource constraints across financial, telecom, and industrial datasets.

Aims and Objectives

Aim:

- To determine whether monitoring configuration is more consequential than detector algorithm under resource-constrained deployment.

Specific Objectives:

1. Compare KS, PSI, and Wasserstein detectors over 8,100 experimental runs.
2. Quantify the effect of monitoring frequency, window size, and reference size on delay and missed detection.
3. Evaluate sensitivity across sudden, gradual, and recurring drift.
4. Assess downstream model robustness (Random Forest, Logistic Regression, LightGBM) under controlled drift.

Related Work

- **Concept drift foundations:** Gama et al. formalize drift types and adaptive monitoring needs.
- **Empirical detector benchmarks:** Barros & Santos; Lukats et al. compare many detectors but do not model deployment constraints as primary factors.
- **Distribution-based monitoring:** KS, PSI, and Wasserstein are practical, label-free, and lightweight for production settings.
- **Open gap addressed in this project:**
  - No prior study systematically ranks resource constraints by their quantitative impact on detection outcomes.

Proposed Solution (Methodology)

- **Datasets:** UCI Credit Default, IBM Telco Churn, and UCI AI4I Predictive Maintenance.
- **Drift simulation:** Sudden, gradual, and recurring drift injected into stream portions after a 60/40 reference-stream split.
- **Detectors:** Kolmogorov-Smirnov test, Population Stability Index, and Wasserstein distance.
- **Constraint grid:**
  - Monitoring frequency: $\{10, 50, 100, 500, 1000\}$
  - Detection window size: $\{50, 100, 500, 1000\}$
  - Reference size: $\{100, 500, \text{unlimited}\}$
- **Analysis:** Kruskal-Wallis, pairwise Mann-Whitney U (Bonferroni), Spearman correlation, and effect sizes.

Key Results

- Constraint configuration produced a **37-fold** range in detection delay, much larger than detector-to-detector differences.
- Gradual drift was significantly harder to detect than sudden/recurring drift ($r \approx 0.30$, medium effect).
- Detection window size had the strongest individual impact; monitoring frequency was second.
- A 500-sample reference window performed comparably to unlimited reference in most settings.
- For post-drift model robustness, Random Forest degraded less than LightGBM ($r = 0.383$) and Logistic Regression ($r = 0.185$).

Conclusion

Main Conclusion

In resource-constrained monitoring, **configuration matters more than algorithm** for practical drift detection performance.

Deployment priority hierarchy:

1. Configure monitoring regime first (frequency, window, reference).
2. Account for drift type behavior, especially gradual drift.
3. Then optimize detector and predictive model selection.

Thank You